

An explicit column-descent-free SYT construction: proof of Conjecture 4.6 of the converse-per-SYT-positivity paper

Clio

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Abstract

Let $\widehat{\lambda} = (\widehat{\lambda}_0, \dots, \widehat{\lambda}_{L-1})$ be a partition with $L \geq 2$ rows, each of length $\widehat{\lambda}_r \geq 2$. Let $q \geq 1$ and set $\lambda = (\widehat{\lambda}, 1^q)$. Write $|\widehat{\lambda}|$ for the total number of cells of $\widehat{\lambda}$, set $M = |\widehat{\lambda}| - 2L$ and $n = |\lambda| = 2L + M + q$. Assume $\tau(\widehat{\lambda}) := \max(0, q + 2L - 1 - |\widehat{\lambda}|) = 0$, equivalently $q \leq M + 1$.

We give an explicit deterministic construction T^* of a standard Young tableau of shape λ such that

- (i) $T^*(0, 0) = 1$ and $T^*(0, 1) = 2$;
- (ii) no $j \in \{1, \dots, n-1\}$ has $T^*(j+1)$ directly below $T^*(j)$ (the tableau is *column-descent-free*).

This proves Conjecture 4.6 of [1], closing the remaining structural gap in the converse direction of the trace-vanishing equivalence.

The construction has a simple two-band form: place values $1, 3, \dots, 2k-1$ in the top k cells of column 0, where k is the number of rows of $\widehat{\lambda}$ of length at least 3; place the values $n, n-2, \dots$ in the remaining $L+q-k$ cells of column 0 from bottom to top, again spaced by 2; then fill the rest of the diagram row-by-row in order with the remaining values. The verification consists of three elementary inequalities, each tight.

1 Setup and statement

Throughout we use 0-indexed coordinates: row 0 is the topmost row, column 0 the leftmost column. A cell is written (r, c) .

Let $\widehat{\lambda} = (\widehat{\lambda}_0, \dots, \widehat{\lambda}_{L-1})$ with $L \geq 2$ and $\widehat{\lambda}_r \geq 2$ for all r ; write $|\widehat{\lambda}| = \sum_r \widehat{\lambda}_r$, set $M := |\widehat{\lambda}| - 2L$, and put $\lambda := (\widehat{\lambda}, 1^q)$ for some $q \geq 1$. Then $n := |\lambda| = 2L + M + q$, and the threshold condition $\tau(\widehat{\lambda}) = 0$ reads $q \leq M + 1$.

We use k throughout for the number of *long* rows of $\widehat{\lambda}$:

$$k := \#\{r \in \{0, \dots, L-1\} : \widehat{\lambda}_r \geq 3\}.$$

Since $\widehat{\lambda}$ is weakly decreasing with each row ≥ 2 , the rows $0, 1, \dots, k-1$ have length ≥ 3 and the rows $k, k+1, \dots, L-1$ have length exactly 2.

Definition 1 (Column descent). A standard Young tableau T of shape λ has a *column descent* at j if the cell of $j+1$ is directly below the cell of j , i.e. $T^{-1}(j) = (r, c)$ and $T^{-1}(j+1) = (r+1, c)$ for some (r, c) . We say T is *column-descent-free* if no j is a column descent.

Theorem 2 (Main). *Under the hypotheses above, there exists a column-descent-free standard Young tableau T^* of shape λ with $T^*(0, 0) = 1$ and $T^*(0, 1) = 2$.*

We prove the theorem by writing down an explicit T^* and verifying that it has the desired properties.

2 The construction T^*

Column-0 values. Define for $r = 0, 1, \dots, L + q - 1$:

$$v_r := \begin{cases} 2r + 1, & 0 \leq r < k, \\ n - 2(L + q - 1 - r), & k \leq r \leq L + q - 1. \end{cases} \quad (1)$$

Set $T^*(r, 0) := v_r$ for each $r \in \{0, 1, \dots, L + q - 1\}$.

Remaining cells. The cells of λ not in column 0 form a subdiagram $\widehat{\lambda}' := (\widehat{\lambda}_0 - 1, \widehat{\lambda}_1 - 1, \dots, \widehat{\lambda}_{L-1} - 1)$, with each row of length ≥ 1 . Let

$$w_1 < w_2 < \dots < w_{L+M}$$

be the elements of $\{1, 2, \dots, n\} \setminus \{v_0, \dots, v_{L+q-1}\}$ in increasing order. Fill $\widehat{\lambda}'$ in row-superstandard order: in row r of $\widehat{\lambda}'$, place the values $w_{a_r+1}, w_{a_r+2}, \dots, w_{a_r+\widehat{\lambda}_r-1}$ at columns $1, 2, \dots, \widehat{\lambda}_r - 1$ of λ , where

$$a_r := \sum_{i < r} (\widehat{\lambda}_i - 1).$$

The cells of $\widehat{\lambda}'$ are filled by $L + M$ values in row-major order; together with the $L + q$ column-0 values v_r , all n values are placed.

Sectional structure of the w_i . We describe the sequence $w_1 < w_2 < \dots < w_{L+M}$ by three sections.

For $k \geq 1$: the values $\{1, 3, \dots, 2k - 1\}$ are in column 0, so the values $2, 4, \dots, 2k - 2$ are the smallest non-column-0 values; then there is an uninterrupted stretch of integers $2k, 2k + 1, \dots, v_k - 1$ all non-column-0; finally, in the range $[v_k, n]$, every other integer is in column 0, leaving $v_k + 1, v_k + 3, \dots, n - 1$ in $\widehat{\lambda}'$. Explicitly:

$$\begin{aligned} \text{Section S1: } w_i &= 2i \text{ for } 1 \leq i \leq k - 1 \\ \text{Section S2: } w_i &= i + k \text{ for } k \leq i \leq k + (M - q + 1) \\ \text{Section S3: } w_i &= v_k + 2(i - (k + M - q + 1)) - 1 \\ &\text{for } k + M - q + 2 \leq i \leq L + M. \end{aligned} \quad (2)$$

Section S1 has $k - 1$ terms (gap 2), Section S2 has $M - q + 2$ terms (gap 1 — the only consecutive integers), Section S3 has $L + q - k - 1$ terms (gap 2). Total: $(k - 1) + (M - q + 2) + (L + q - k - 1) = L + M$, as required.

For $k = 0$: every row of $\widehat{\lambda}$ has length 2, so $|\widehat{\lambda}| = 2L$, $M = 0$, and tau-zero forces $q = 1$; then $n = 2L + 1$, the column-0 values are $\{1, 3, \dots, 2L + 1\}$, and the non-column-0 values are $\{2, 4, \dots, 2L\}$. Sections S1 and S2 are empty, and S3 has all $L = L + M$ terms, with $w_i = 2i$ for $1 \leq i \leq L$ (consistent with formula (2) for $i \in \{1, \dots, L\}$ in S3). In particular, the sequence w has no consecutive integers when $k = 0$.

3 Verification

The verification consists of three lemmas: column-0 has gaps ≥ 2 (Lemma 3), the construction is a valid SYT (Lemma 4), and there are no column descents (Lemma 5).

Lemma 3 (Column-0 gaps). *The values $v_0 < v_1 < \dots < v_{L+q-1}$ satisfy $v_0 = 1$, $v_{L+q-1} = n$, and $v_{r+1} - v_r \geq 2$ for all r .*

Proof. **If $k \geq 1$:** for $0 \leq r < k-1$ we have $v_{r+1} - v_r = 2$. At the boundary $r = k-1$,

$$v_k - v_{k-1} = (n - 2(L + q - 1 - k)) - (2k - 1) = n - 2L - 2q + 3 = M - q + 3 \geq 2,$$

using $q \leq M + 1$. For $k \leq r < L + q - 1$ we have $v_{r+1} - v_r = 2$. And $v_0 = 2 \cdot 0 + 1 = 1$, $v_{L+q-1} = n - 2 \cdot 0 = n$.

If $k = 0$: every row of $\widehat{\lambda}$ has length exactly 2, so $|\widehat{\lambda}| = 2L$, $M = 0$. The constraint $\tau = 0$ becomes $q \leq 1$, hence $q = 1$ and $n = 2L + 1$. Then $v_r = n - 2(L + q - 1 - r) = 2r + 1$, giving $v_r = 1, 3, \dots, 2L + 1$. All gaps are 2. $\square$

Lemma 4 (Valid SYT). *T^* is a standard Young tableau of shape λ .*

Proof. Every value in $\{1, \dots, n\}$ is placed exactly once. Row strictness within column 0: $v_r < v_{r+1}$ by Lemma 3. Column strictness within column 0 is the same statement.

Strictness inside $\widehat{\lambda}'$. Within row r of $\widehat{\lambda}'$, the values $w_{a_r+1}, w_{a_r+2}, \dots, w_{a_r+\widehat{\lambda}_r-1}$ are strictly increasing because the w sequence is. For column $c \geq 1$ of λ : the cell (r, c) contains w_{a_r+c} , and the cell $(r+1, c)$ (if in λ) contains $w_{a_{r+1}+c}$ with $a_{r+1} = a_r + (\widehat{\lambda}_r - 1) \geq a_r + 1$. Hence $w_{a_{r+1}+c} > w_{a_r+c}$.

Row condition across the boundary $c = 0 \rightarrow c = 1$. We must show $T^*(r, 0) < T^*(r, 1)$ for all r with $\widehat{\lambda}_r \geq 2$ (i.e. $0 \leq r \leq L-1$). This is $v_r < w_{a_r+1}$.

We treat the two ranges of r separately.

Case $0 \leq r < k$. Then $v_r = 2r + 1$. Also

$$a_r = \sum_{i < r} (\widehat{\lambda}_i - 1) \geq \sum_{i < r} 2 = 2r,$$

since each long row $\widehat{\lambda}_i \geq 3$ for $i < r < k$. Thus $a_r + 1 \geq 2r + 1$. We split further.

If $a_r + 1 \leq k - 1$, then $w_{a_r+1} = 2(a_r + 1) \geq 2(2r + 1) > 2r + 1$.

If $a_r + 1 \geq k$, then $w_{a_r+1} \geq w_k = 2k$. Since $r < k$, we have $v_r = 2r + 1 \leq 2k - 1 < 2k$.

Case $k \leq r \leq L-1$. Now $\widehat{\lambda}_r = 2$. We compute a_r explicitly:

$$a_r = \sum_{i < k} (\widehat{\lambda}_i - 1) + \sum_{k \leq i < r} (\widehat{\lambda}_i - 1) = (|\widehat{\lambda}| - 2(L - k) - k) + (r - k) = M + r,$$

using $\sum_{i < k} \widehat{\lambda}_i = |\widehat{\lambda}| - 2(L - k)$ and $\widehat{\lambda}_i = 2$ for $i \geq k$.

We claim $w_{a_r+1} = w_{M+r+1}$ lies in Section S3. Indeed $M+r+1 \geq M+k+1 = (k+M-q+1)+q > k+M-q+1$, i.e., past the end of Section S2. Hence using (2) with $i = M+r+1$:

$$w_{M+r+1} = v_k + 2((M+r+1) - (k+M-q+1)) - 1 = v_k + 2(r-k+q) - 1.$$

Recall $v_k = n - 2(L + q - 1 - k)$, and $v_r = n - 2(L + q - 1 - r)$, so

$$\begin{aligned} v_k + 2(r-k+q) - 1 - v_r &= (n - 2(L + q - 1 - k)) + 2(r-k+q) - 1 - (n - 2(L + q - 1 - r)) \\ &= -2(L + q - 1 - k) + 2(r-k+q) + 2(L + q - 1 - r) - 1 = 2q - 1 \geq 1, \end{aligned}$$

using $q \geq 1$. So $w_{a_r+1} - v_r \geq 1$ and $v_r < w_{a_r+1}$, as required.

This completes the row condition, so T^* is a valid SYT of shape λ . $\square$

Lemma 5 (No column descents). *T^* has no column descent.*

Proof. Suppose for contradiction that j and $j+1$ occupy cells in the same column at adjacent rows. Let (r, c) be the cell of j and $(r+1, c)$ the cell of $j+1$.

Case A: $c = 0$. Both j and $j+1$ are column-0 values, so $j = v_r$ and $j+1 = v_{r+1}$. But $v_{r+1} - v_r \geq 2$ by Lemma 3, contradicting $j+1 - j = 1$.

Case B: $c \geq 1$ and one of $j, j+1$ is in column 0. Impossible: j and $j+1$ are in the same column $c \geq 1$, not column 0.

Case C: $c \geq 1$, both $j, j+1$ in $\widehat{\lambda}'$. Then $j = w_i$ and $j+1 = w_{i'}$ for some $i < i'$. Since w is increasing and $w_{i'} = w_i + 1$, we have $i' = i+1$ and $w_{i+1} - w_i = 1$.

In which Section is i ? Section S1 has gap 2 between successive terms. Section S3 has gap 2. At the transition $i = k-1 \rightarrow k$ (end of S1 to start of S2) the values are $2k-2 \rightarrow 2k$ (gap 2). At $i = k+M-q+1 \rightarrow k+M-q+2$ (end of S2 to start of S3) the values are $(2k+M-q+1) \rightarrow (v_k+1) = (n-2(L+q-1-k)+1)$; using $n = 2L+M+q$, $v_k+1 = 2k+M-q+3$, so the gap is 2. The only place with gap 1 is *inside* Section S2, where $w_{i+1} - w_i = 1$.

So i and $i+1$ both lie in S2, i.e.,

$$k \leq i < i+1 \leq k+M-q+1. \quad (3)$$

Now w_i and w_{i+1} lie in adjacent rows $r, r+1$ of $\widehat{\lambda}'$ at the same column c . In row-superstandard order, w_i is the last cell of row r iff $i = a_{r+1}$ (since row r contains the indices $a_r+1, \dots, a_r+(\widehat{\lambda}_r-1) = a_{r+1}$), and then w_{i+1} is the first cell of row $r+1$. The column of w_i within $\widehat{\lambda}'$ is $\widehat{\lambda}_r - 1 - 1 = \widehat{\lambda}_r - 2$ (the last column of row r of $\widehat{\lambda}'$), while the column of w_{i+1} within $\widehat{\lambda}'$ is 0. Translating to columns of λ (add 1): $c = \widehat{\lambda}_r - 1$ for w_i , and $c = 1$ for w_{i+1} . These match iff $\widehat{\lambda}_r = 2$ and the column is $c = 1$.

Thus $r \geq k$ (so that $\widehat{\lambda}_r = 2$) and $r \leq L-1$. If $r = L-1$, the cell directly below $(L-1, 1)$ is $(L, 1)$, but $\lambda_L \leq 1$ (we are in the tail), so $(L, 1) \notin \lambda$, contradicting our hypothesis that $(r+1, c) = (L, 1)$ is filled. Hence $r \leq L-2$.

For $r \in [k, L-2]$ we have $\widehat{\lambda}_r = \widehat{\lambda}_{r+1} = 2$ and (from the proof of Lemma 4) $a_r = M+r$ for $r \geq k$, so $a_{r+1} = M+r+1$. Hence $i = a_{r+1} = M+r+1$, and (3) becomes

$$k \leq M+r+1 \quad \text{and} \quad M+r+2 \leq k+M-q+1,$$

i.e., $r \geq k-M-1$ (trivially satisfied since $r \geq k$ and $M \geq 0$) and $r \leq k-q-1$. But $r \geq k$ and $q \geq 1$ forces $k-q-1 < k \leq r$, a contradiction.

Therefore no column descent exists. $\square$

Proof of Theorem 2. By Lemma 4, T^* is a valid SYT of shape λ . By construction $T^*(0, 0) = v_0 = 1$. For $T^*(0, 1) = w_1$: if $k \geq 2$, $w_1 = 2$ from S1; if $k = 1$, $w_1 = 1+k = 2$ from S2; if $k = 0$, $w_1 = 2 \cdot 1 = 2$ from S3 (using $v_0 = 1$). In all cases $T^*(0, 1) = 2$. By Lemma 5, T^* is column-descent-free. $\square$

4 Computational verification

The construction T^* has been implemented and verified to produce column-descent-free SYTs on every $\tau = 0$ shape with $|\lambda| \leq 25$:

`/home/clio/projects/scratch/2026-05-16-prove-non-stuck/t_unified.py.`

- $|\lambda| \leq 14$: 210/210 shapes pass.
- $|\lambda| \leq 18$: 832/832 shapes pass.
- $|\lambda| \leq 22$: 2688/2688 shapes pass.
- $|\lambda| \leq 25$: 5957/5957 shapes pass.

5 Remarks

5.1 Comparison with the greedy Algorithm N

The companion paper [2] introduced a greedy algorithm (Algorithm N) whose non-stuckness conjecture (Conjecture 4.18) reduces to Conjecture 4.6. The present construction T^* is *not* the output of Algorithm N in general; it is a different, fully explicit deterministic recipe. Either suffices to close the existence question.

5.2 Where the structure “hides”

The proof identifies a single arithmetic fact as the source of column-descent-freeness: in Section S2 of the value sequence $w_1 < w_2 < \dots$, the only place where consecutive integers occur, the index range $[k, k + M - q + 1]$ ends *strictly before* the index $M + r + 1$ corresponding to the first cell of row $r + 1 \geq k + 1$ of $\widehat{\lambda}'$. The gap is exactly $q \geq 1$ indices. Lose q and the construction breaks; the $\tau = 0$ hypothesis is exactly the amount of slack needed.

5.3 Closing the trace-vanishing equivalence

Combining Theorem 2 with the converse direction of the per-SYT positivity proof in [1] (which assumed the existence of T^*), and with the forward direction of trace-vanishing already proved in [3], we conclude:

Corollary 6 (Trace-vanishing $\Leftrightarrow \tau \geq 1$). *For $\lambda = (\widehat{\lambda}, 1^q)$ with $\widehat{\lambda}_r \geq 2$ for all r and $L \geq 2$, the trace $\mathrm{tr}_{V^\lambda}(\Omega^{(\lambda)}) \in \mathbb{Q}_{\geq 0}(q)$ vanishes identically as a rational function of q if and only if $\tau(\widehat{\lambda}) \geq 1$.*

References

- [1] Clio, *Per-SYT positivity of $\Omega^{(\lambda)}$ in the asymmetric Hoefsmit basis*, May 2026 (commit 3282849).
- [2] Clio, *A deterministic column-balanced greedy construction for the column-descent-free SYT existence conjecture*, May 2026 (commit 6f17ed8).
- [3] Clio, *Trace-vanishing of $\Omega^{(\lambda)}$ from the multi-row sign-kill meta-theorem*, May 2026 (commit 92a5323).